

Deployment Strategies

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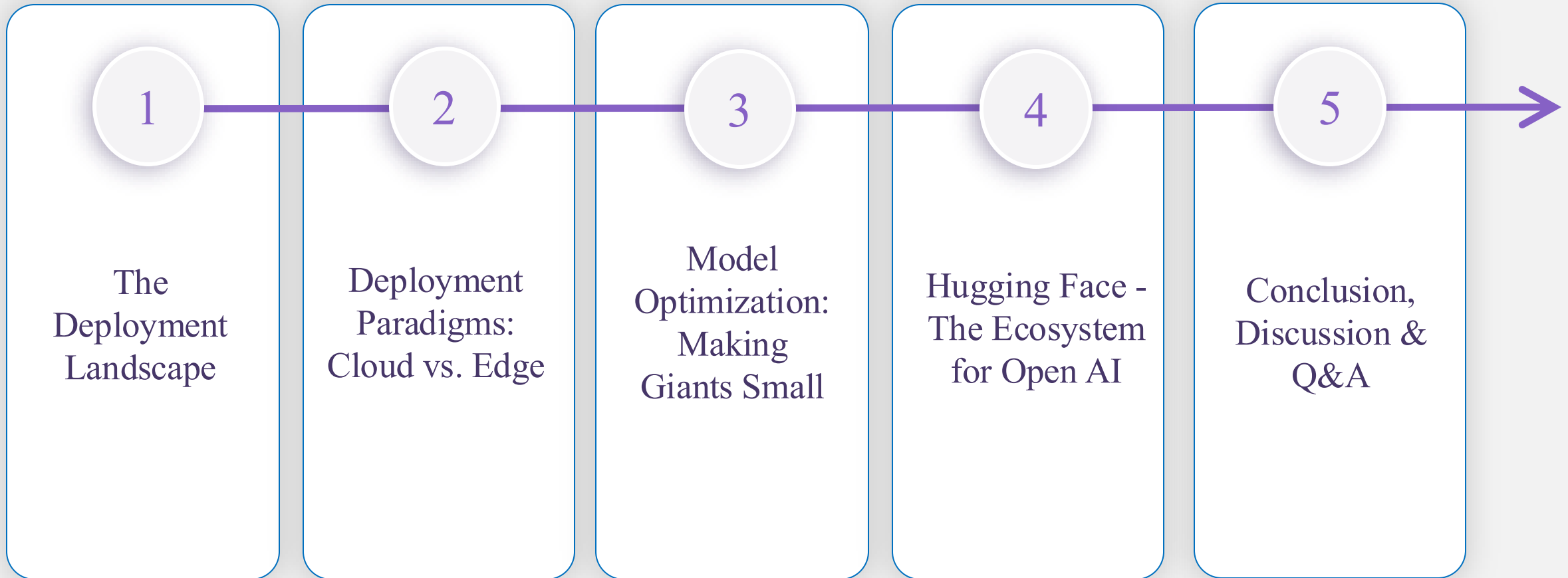
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Orlando, November 17 – 25, 2025



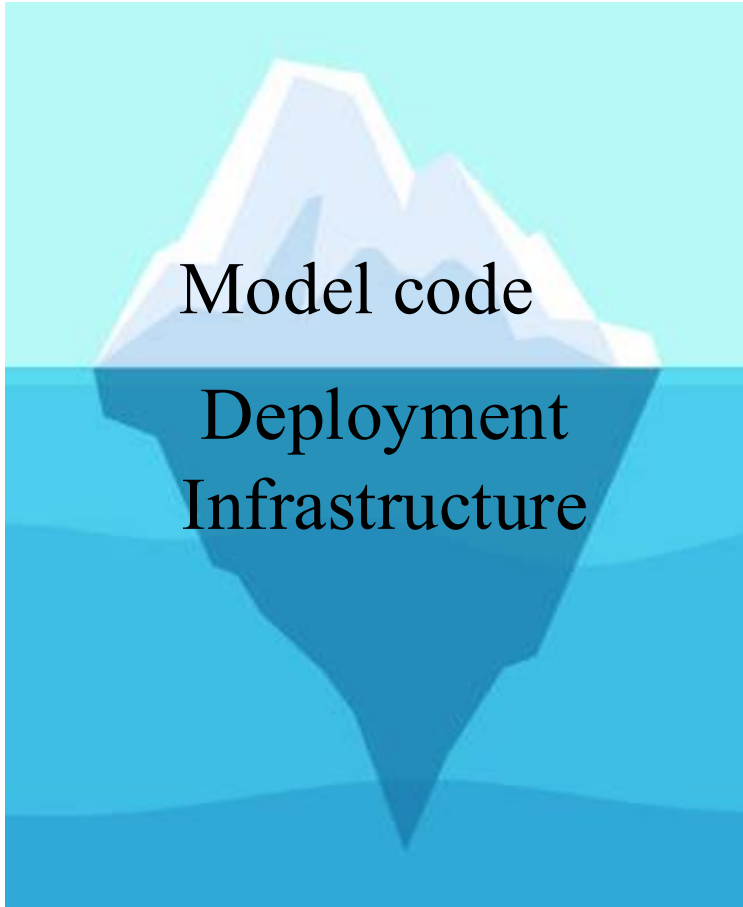
*This activity
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Agenda



The AI "Valley of Death"



The "Valley of Death": The gap between a working prototype (Jupyter Notebook) and an operational system.

In disaster management, "production" doesn't mean a smooth web app; it means a tool a first responder can trust in the mud and rain.

Research vs. Production: The Mindset Shift

Feature	Research (The Notebook)	Production (The Field)
Data	Static, Cleaned, CSV/Zip	Dynamic, Streaming, Noisy, Missing
Goal	Maximize Accuracy (F1-Score)	Reliability & Latency
Hardware	Infinite GPU (Colab/Cluster)	Constrained (Battery, CPU, Edge)
Success	A Paper / Leaderboard	An Actionable Decision

The "Colab Illusion"



The Environment Trap: We have spent the last week in a perfect environment.

- Pre-installed libraries (pip install works instantly).
- High-speed internet to download weights.
- Tesla T4 or V100 GPUs provided for free.

The Real World:

- No internet to download libraries.
- No sudo/admin rights on the command computer.
- Hardware is a ruggedized laptop from 2019 or an embedded Jetson board.

"It works on my machine" is not a valid defense in operational scenarios.

The Disaster Context: The "DDIL" Environment



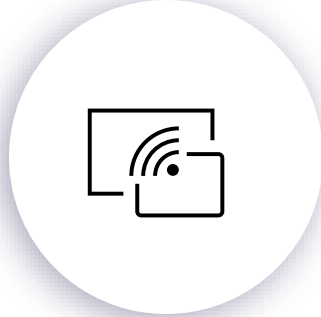
Denied: No connectivity allowed (security protocols) or available (infrastructure destroyed).

Degraded: 2G speeds, packet loss, high latency.

Intermittent: Connection drops every few minutes.

Limited: Bandwidth constraints (sharing satellite link with entire command post).

The Unique Constraints of Disaster Management



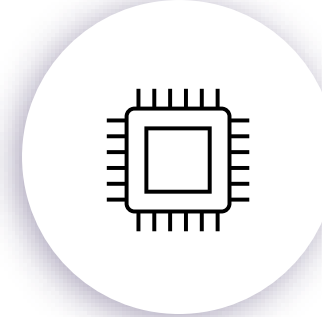
Connectivity

Often
intermittent or
non-existent
(requires offline
capability)



Latency

"Golden hours"
decisions need to be
made in seconds,
not hours



Power/Compute

Running on battery-
powered drones or
handheld devices
vs. massive cloud
servers

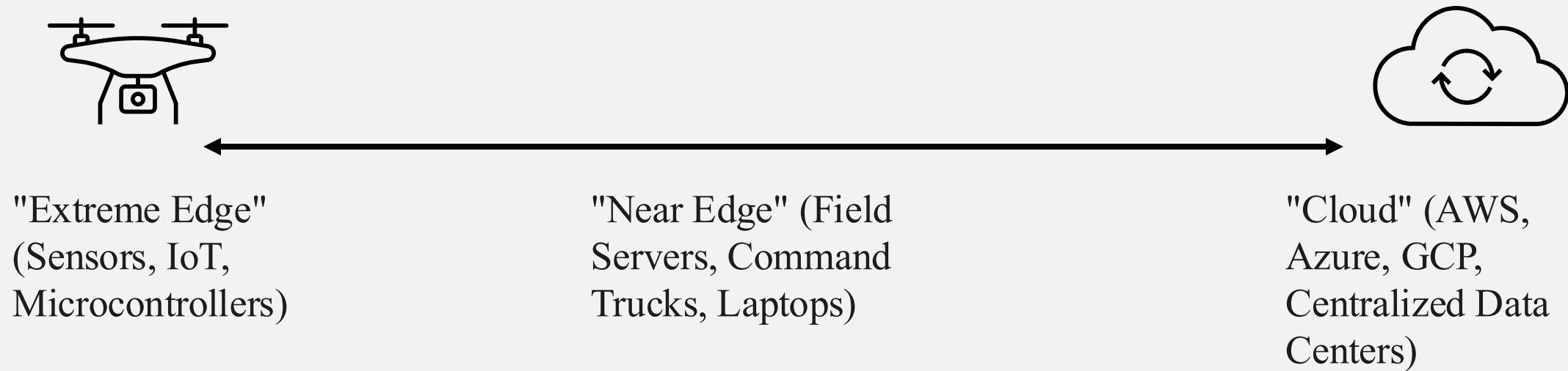
The "It Works on My Machine" Fallacy



A model with **99% accuracy** is **useless** if it takes 10 minutes to process one image or requires an **internet connection** that doesn't exist.

Deployment is the process of **integrating a ML/DL model into an existing production environment** to make practical business decisions based on data.

Deployment Paradigms: Cloud vs. Edge



Deployment is not a binary choice; it is a spectrum of compute availability vs. latency requirements.

Key Question: Where does the data live, and where does the intelligence live?

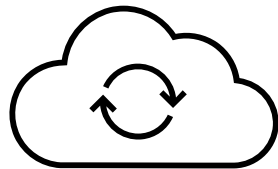
Cloud vs. Edge Deployment

Cloud Deployment (Centralized):

Pros: Unlimited compute, easy to scale, centralized updates.

Cons: High latency, requires stable internet, privacy concerns.

Use Case: Post-event analysis of satellite imagery for damage assessment reports.

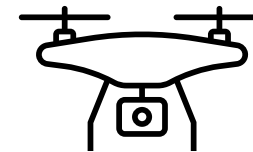


Edge Deployment (Decentralized):

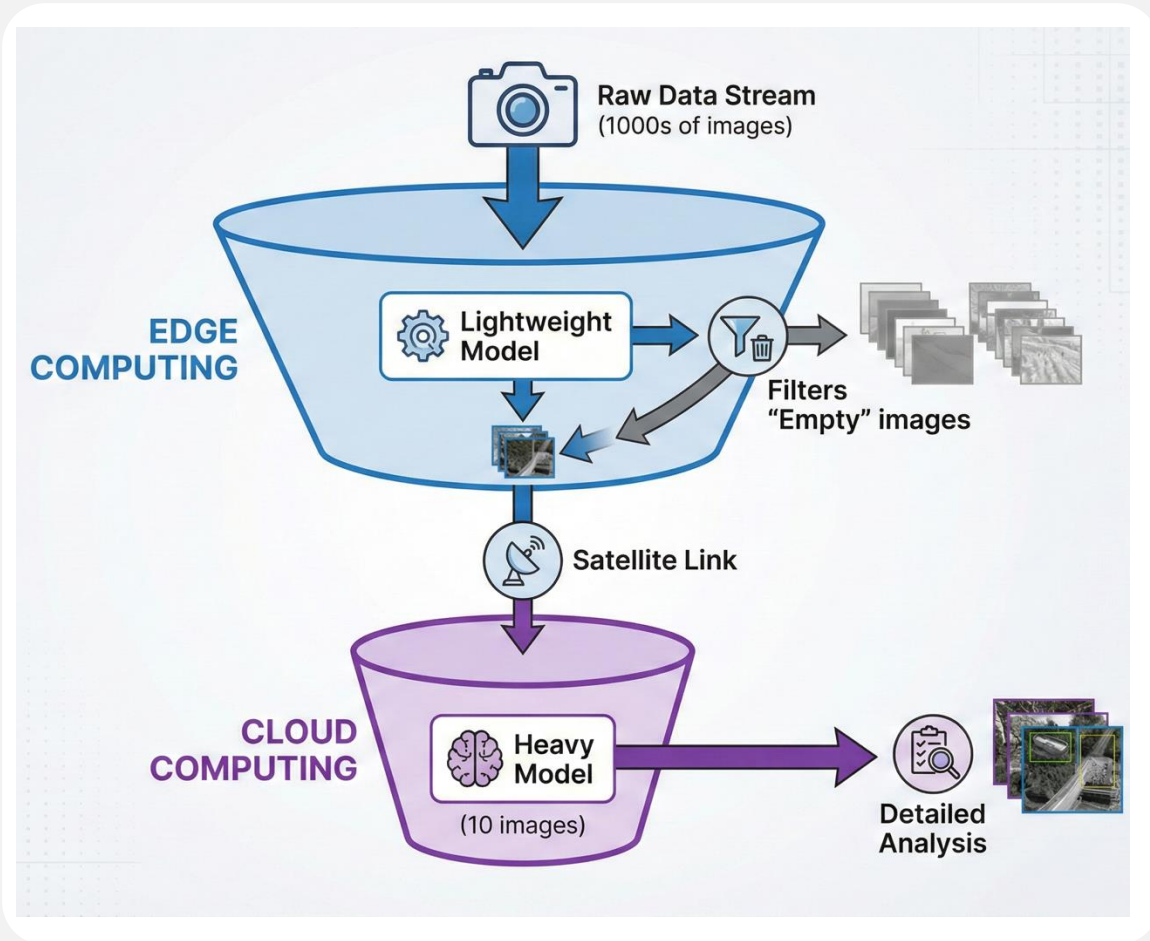
Pros: Real-time, no internet needed, data stays on device.

Cons: Limited compute (battery/heat), harder to update models.

Use Case: A drone identifying survivors or flood boundaries in real-time.



Hybrid Approach (The "Intelligent Triage")



Step 1 (Edge): Drone flies and uses a small model to detect potential damage.

Step 2 (Filter): It discards images of forests/water (90% of data).

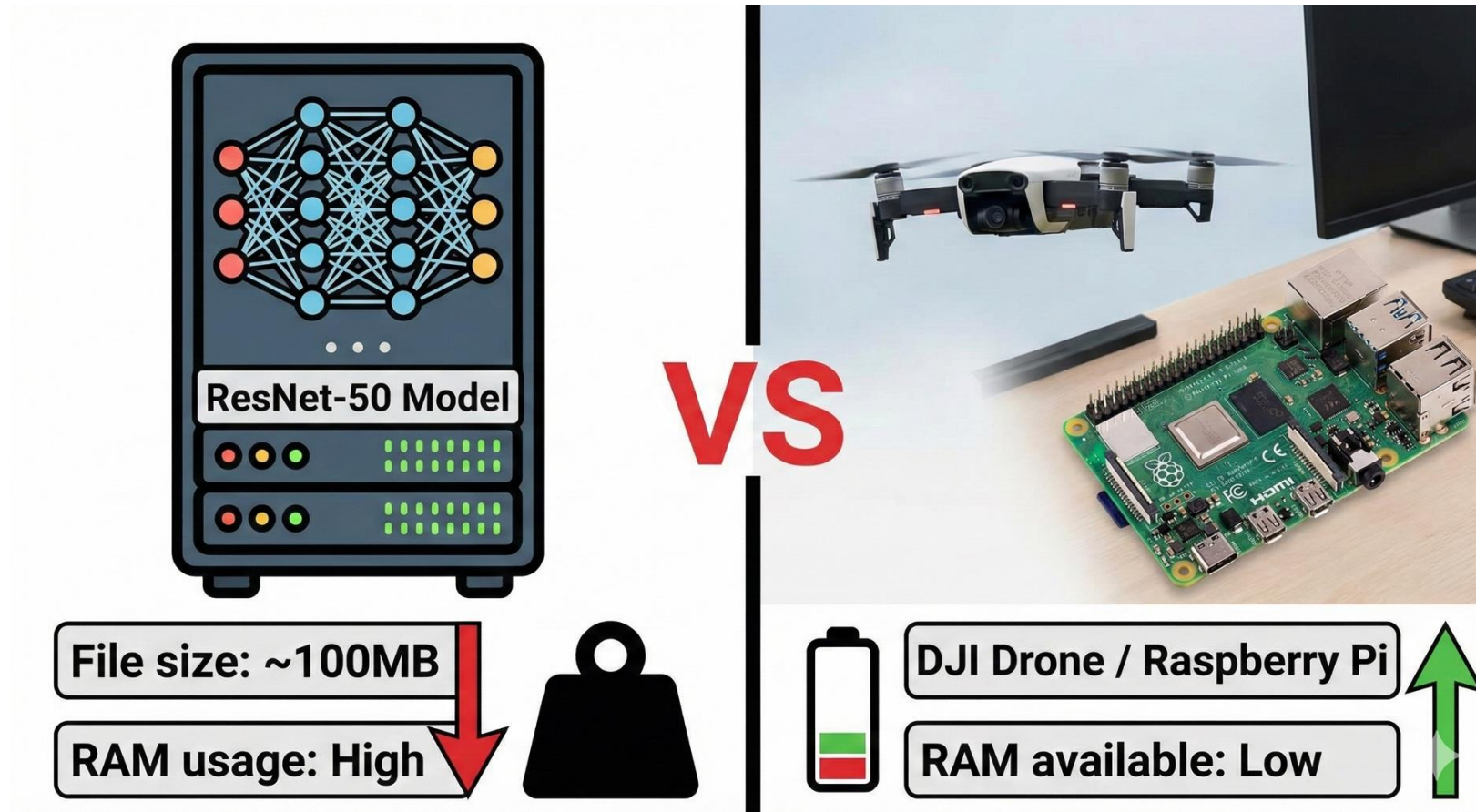
Step 3 (Transmit): It only uploads the 10% of images containing buildings/roads.

Step 4 (Cloud): Cloud server runs a high-precision segmentation model on the flagged images.

Summary Matrix: Choosing Your Strategy

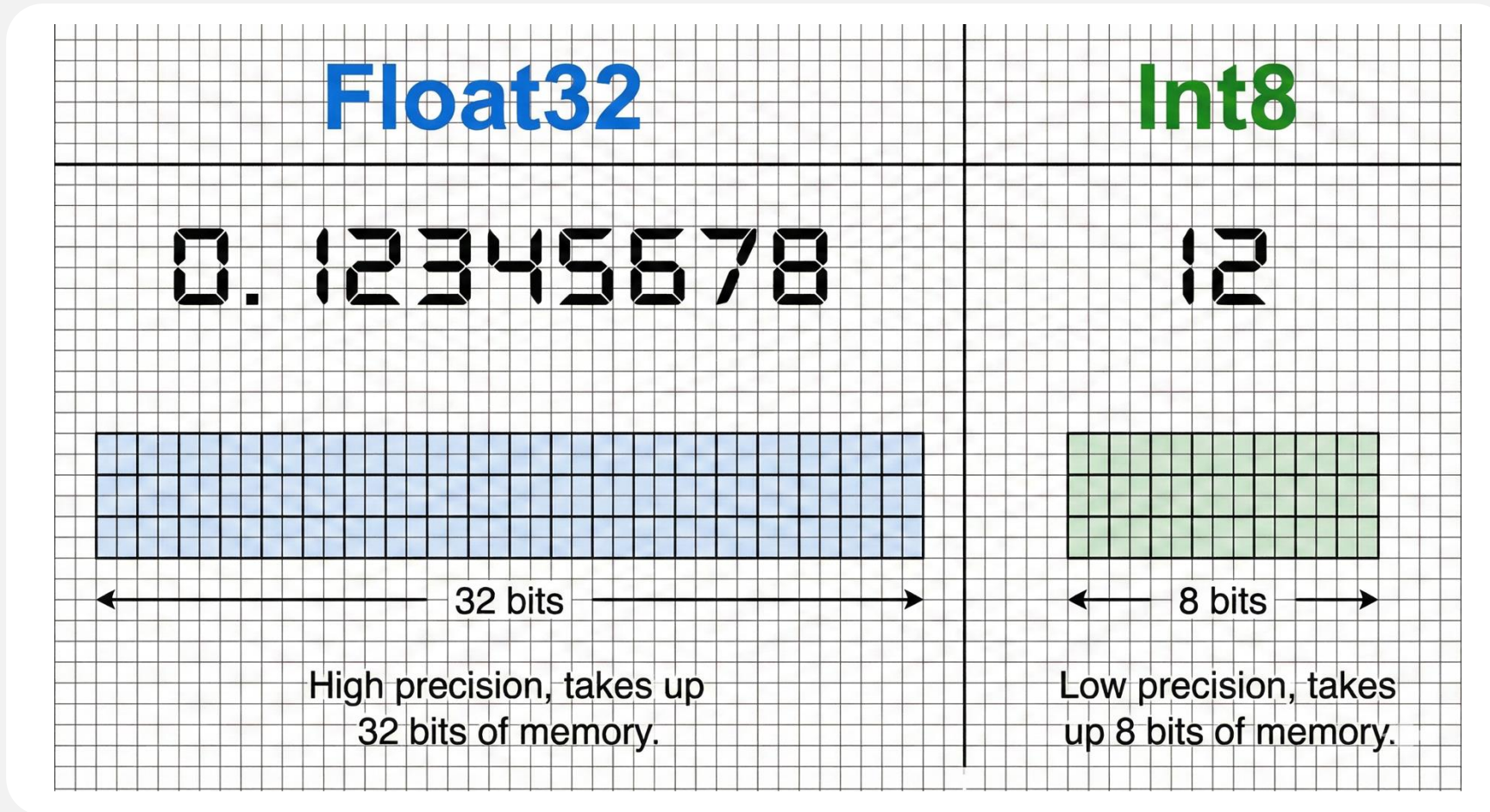
Feature	Cloud	Edge	Hybrid
Compute Power	High	Low	Medium
Latency	High	Low	Variable
Bandwidth Need	High	None (or very low)	Low
Primary Goal	Accuracy	Speed	Efficiency
Ideal for...	Strategic Planning	First Response	Long-term Ops

Model Optimization



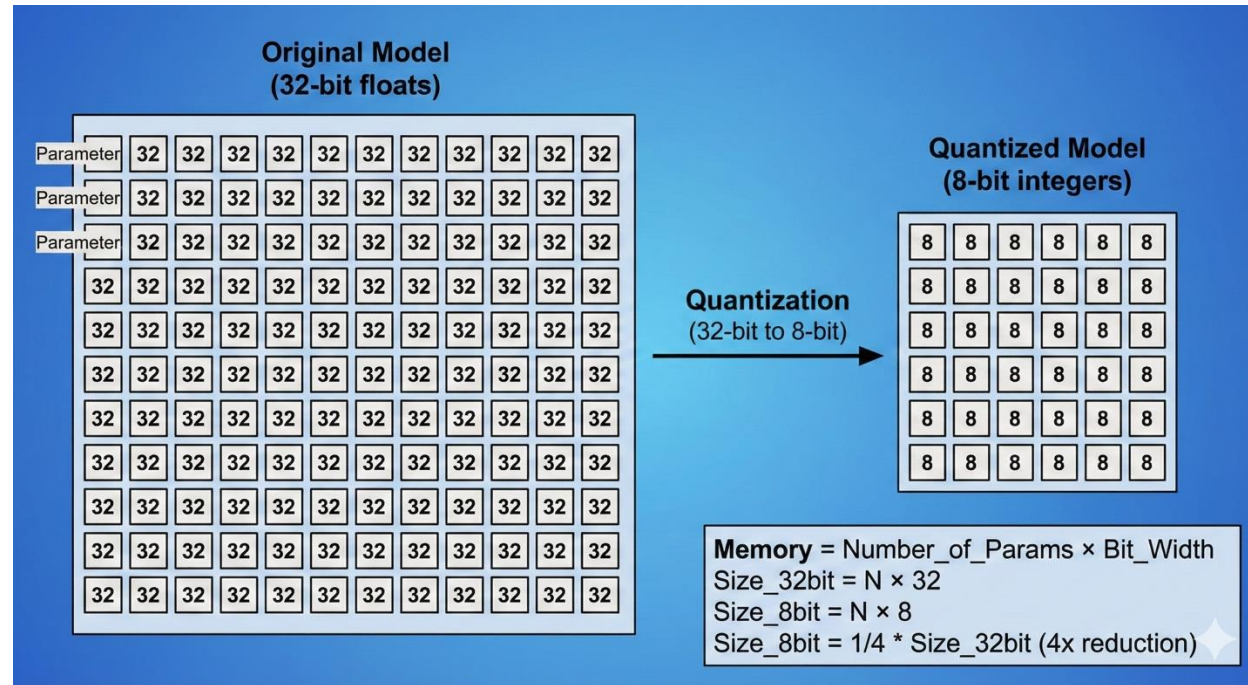
Deep Learning models are over-parameterized giants trying to fit into tiny homes.

Technique 1: Quantization (The Low-Hanging Fruit)



Converting the model's weights and activations from high-precision floating-point numbers (32-bit) to lower-precision integers (8-bit).

Technique 1: Quantization (The Low-Hanging Fruit)



The Trade-off:

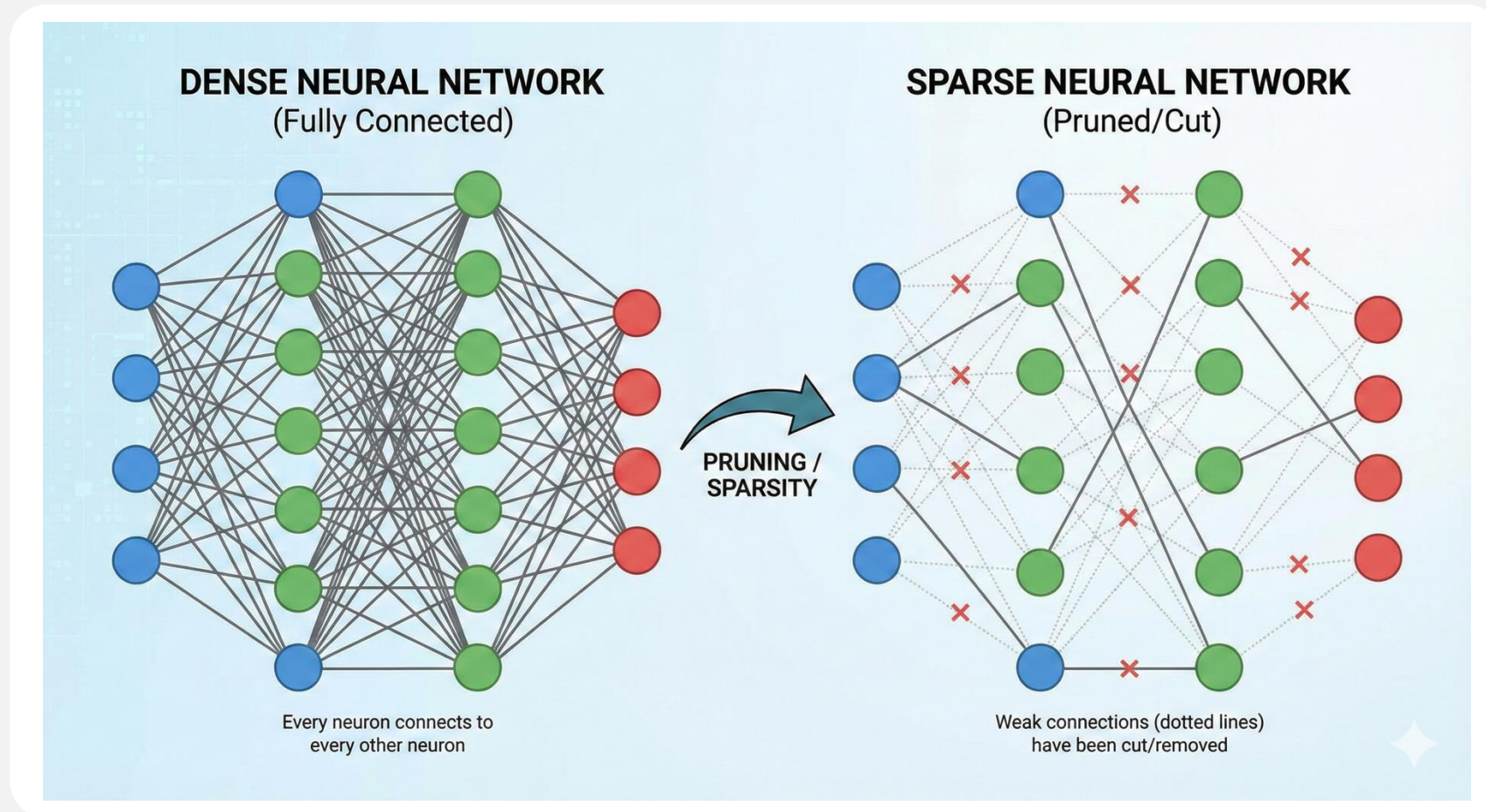
Size: 75% smaller.

Speed: 2x-4x faster (processors handle integers faster than floats).

Accuracy: Usually negligible loss (<1%).

Disaster Context: A drone doesn't need to know the probability of "Flood" is 99.12345%; it just needs to know it's >90%.

Technique 2: Pruning (Sparse Networks)



Neural networks are "lazy." After training, many neurons have weights close to zero or redundant contributions.

Technique 2: Pruning (Sparse Networks)

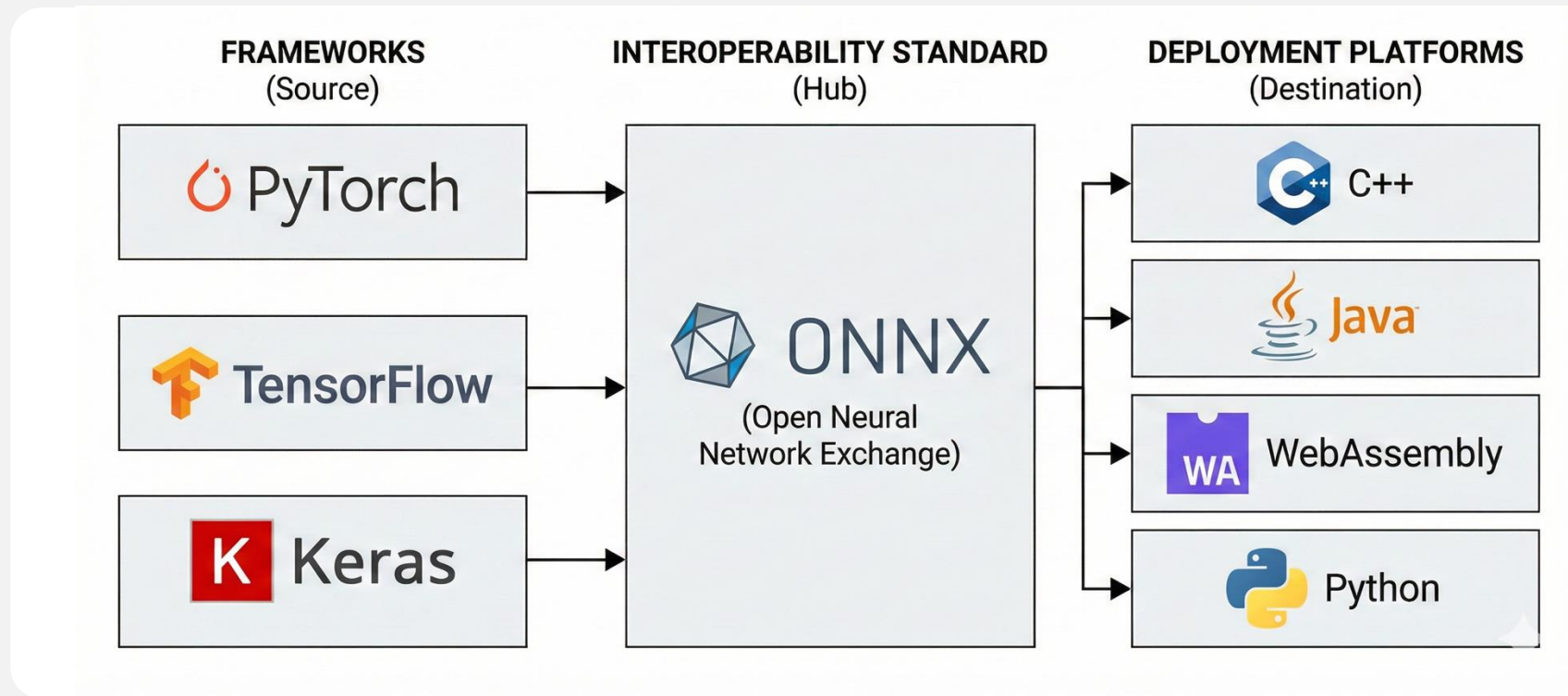
The Process:

1. Train the model.
2. Identify "unimportant" connections (weights near zero).
3. Delete them (set to exactly zero).
4. Fine-tune (retrain slightly) to heal the damage.

The Result: A model that is 50-90% sparse (mostly zeros).

The Catch: You often need specialized hardware or libraries to see the speedup (standard GPUs hate sparse matrices).

Interoperability: The "Universal Translator" (ONNX)



The Problem: You train in Python/PyTorch, but the drone's flight controller runs in C++. Installing PyTorch on a drone is a nightmare.

The Solution (ONNX): An open standard format to represent ML models.

Interoperability: The "Universal Translator" (ONNX)

Workflow:

1. `torch.onnx.export(model)` → Creates a .onnx file.
2. This file is static. It describes the math, not the code.
3. Load this file in any language (C#, Java, C++) using ONNX Runtime.

Benefit: Write in Python, Deploy Anywhere.

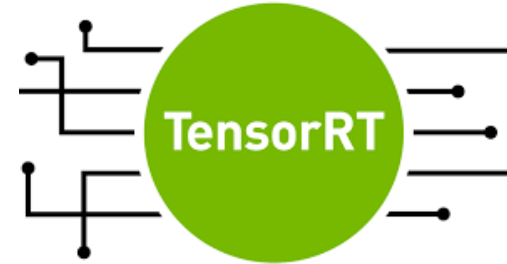
Specialized Runtimes (The Right Engine for the Car)



TensorFlow Lite / Lite Micro

Target: Mobile devices (Android/iOS), Raspberry Pi, Microcontrollers.

Magic: Optimized for ARM processors and low power.



NVIDIA TensorRT:

Target: NVIDIA GPUs (Server or Jetson Edge devices).

Magic: Layer Fusion.
(Combines multiple math operations into one step to reduce memory access).

Hugging Face - The Ecosystem for Open AI



Hugging Face

The "GitHub of AI"

- The world's largest open-source community and platform for ML/DL.
- A central hub where researchers share Models, Datasets, and Demos (Spaces).
- **Goal:** To democratize good machine learning, making SOTA accessible to everyone, not just tech giants.

Hugging Face - The Ecosystem for Open AI

AI is no longer confined to research labs; it's a tool that anyone can use to build intelligent, interactive real-world applications.

Hugging Face allows us to use Transfer Learning - taking a model that already understands satellite imagery and fine-tuning it for our specific flood or earthquake data.

Hugging Face - The Ecosystem for Open AI



Models

huggingface.co/models

Hosting over 1,000,000+ pre-trained models

Relevance: Instead of training a damage detection model from scratch (weeks of compute), you can download a ViT pre-trained on satellite imagery in seconds



Datasets

huggingface.co/datasets

A library of hosted, accessible datasets

Relevance: This is where to host specific datasets



Easy-to-build web applications to demonstrate your model's capabilities to stakeholders without needing a frontend developer

Hugging Face Models

 **Hugging Face**

Models

Datasets

Spaces

Community

Docs

Enterprise

Pricing

Main

Tasks

Libraries

Languages

Licenses

Other

Tasks

Text Generation

Any-to-Any

Image-Text-to-Text

Image-to-Text

Image-to-Image

Text-to-Image

Text-to-Video

Text-to-Speech

+ 42

Parameters

< 1B

6B

12B

32B

128B

> 500B

Libraries

PyTorch

TensorFlow

JAX

Transformers

Diffusers

sentence-transformers

Safetensors

ONNX

GGUF

Transformers.js

MLX

MLX

+ 41

Apps

vLLM

TGI

llama.cpp

MLX LM

LM Studio

Ollama

Jan

+ 13

Inference Providers

Groq

Novita

Nebius AI

Cerebras

SambaNova

Nscale

fal

Hyperbolic

+ 11

Models

2,240,534

Filter by name

Full-text search

Inference Available

Sort: Trending

tencent/HunyuanVideo-1.5

Text-to-Video • Updated 2 days ago • 243 • 487

WeiboAI/VibeThinker-1.5B

Text Generation • 2B • Updated 8 days ago • 14k • 436

moonshotai/Kimi-K2-Thinking

Text Generation • Updated 15 days ago • 218k • 1.36k

maya-research/maya1

Text-to-Speech • 3B • Updated 11 days ago • 38.7k • 729

dx8152/Qwen-Edit-2509-Multiple-angles

Image-to-Image • Updated 11 days ago • 81.8k • 724

deepseek-ai/DeepSeek-OCR

Image-Text-to-Text • 3B • Updated 19 days ago • 5.1M • 2.81k

miromind-ai/MiroThinker-v1.0-72B

Text Generation • 73B • Updated 4 days ago • 2.23k • 120

nvidia/NVIDIA-Nemotron-Parse-v1.1

Image-Text-to-Text • Updated 3 days ago • 2.48k • 74

PleIAs/Baguettotron

Text Generation • 0.3B • Updated 5 days ago • 6.77k • 194

facebook/sam3

Mask Generation • 0.9B • Updated 2 days ago • 50.4k • 482

Supertone/supertonic

Text-to-Speech • Updated 1 day ago • 594 • 163

facebook/sam-3d-objects

Updated 1 day ago • 135

dx8152/Qwen-Edit-2509-Multi-Angle-Lighting

Image-to-Image • Updated 2 days ago • 2.28k • 125

eigen-ai-labs/eigen-banana-qwen-image-edit

Text-to-Image • Updated 7 days ago • 398 • 208

salakash/SamKash-Tolstoy

Text Generation • Updated about 7 hours ago • 3.05k • 306

Phr00t/Qwen-Image-Edit-Rapid-AIO

Text-to-Image • Updated 3 days ago • 795

allenai/Olmo-3-32B-Think

1.05M • Updated 1 day ago • 1.05k • 74

baidu/ERNIE-4.5-VL-28B-A3B-Thinking

Image-Text-to-Text • 30B • Updated 9 days ago • 37.5k • 492

Hugging Face Workflow

Accelerating Disaster Response: From Problem to Model

STEP 1: Define the Mission

"We have raw aerial imagery from a drone flight over a flood zone. We need to identify which roads are impassable."

The Data: Real-world, noisy, unstructured data (Satellite GeoTIFFs, Drone Video, Social Media Text)

STEP 2: Model Hub Discovery

Task: Image Segmentation (for floods), Object Detection (for damaged buildings).

Libraries: PyTorch, TensorFlow.

Datasets: Search for models fine-tuned on datasets like xView2 or BigEarthNet.

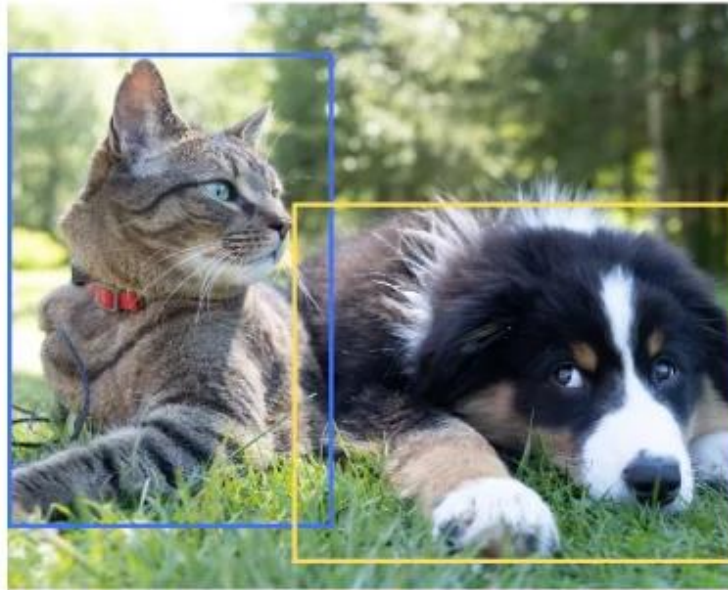
Hugging Face Models (Tasks)

Is this a cat?



Image Classification

What is there in the image
and where?



Object Detection

Which pixels belong to
which object



Image Segmentation

<https://www.superannotate.com/blog/image-segmentation-for-machine-learning>

Hugging Face Workflow

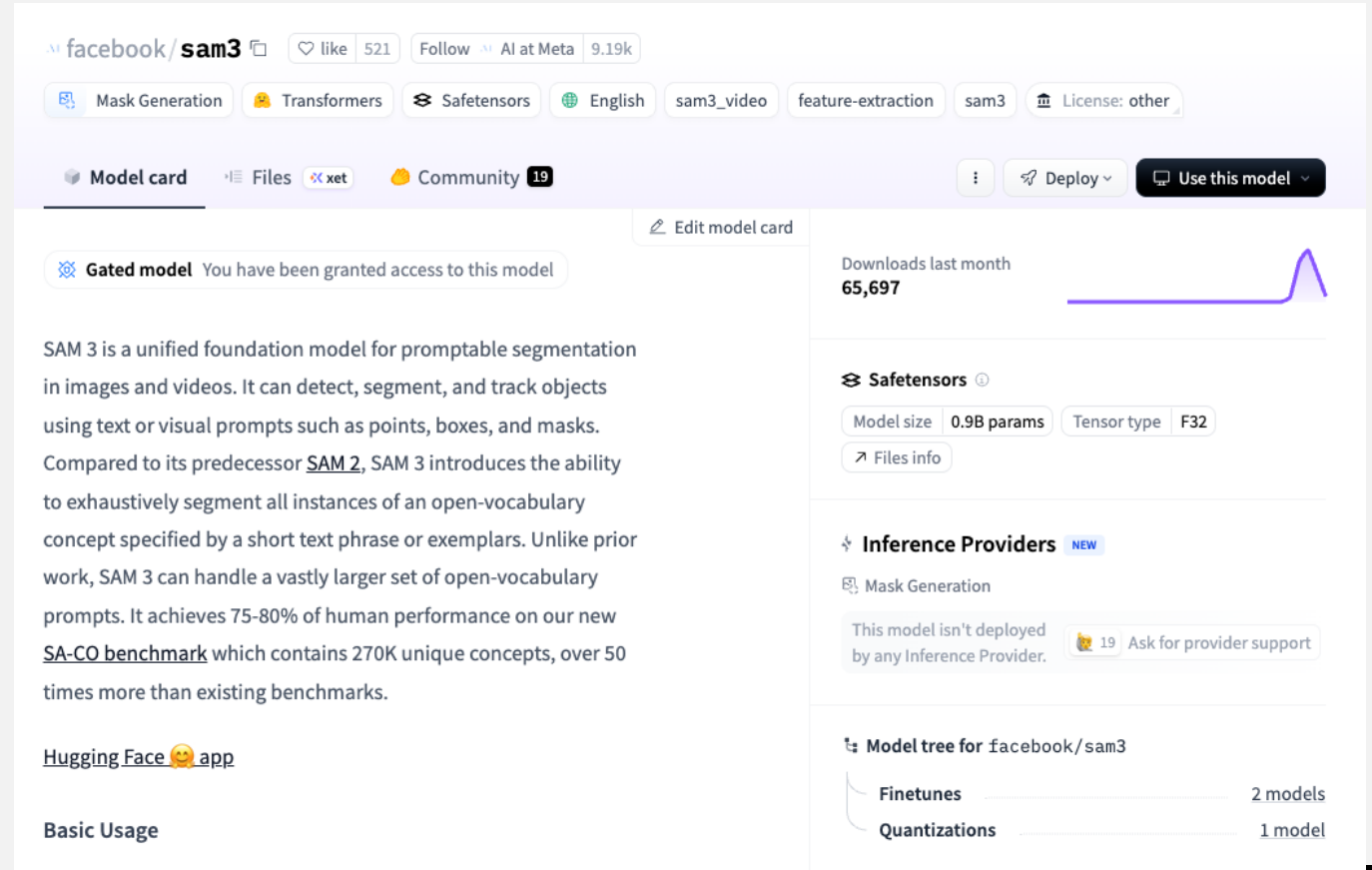
Accelerating Disaster Response: From Problem to Model

STEP 3: The Model Card

Training Data: Did this model see diverse geographies? (e.g., A model trained only on US suburbs may fail in rural diverse regions).

Benchmarks: Precision vs. Recall (Do we prioritize finding all damaged buildings, or avoiding false alarms?).

Usage: Copy-paste snippets to test the pipeline immediately.



Hugging Face Workflow

Accelerating Disaster Response: From Problem to Model

STEP 4: Path A Hosted Inference API

Best For: Rapid prototyping, post-event analysis in connected command centers.

Mechanism: Send imagery via HTTP POST request → Hugging Face GPU Cluster processes it → Return JSON.

Pros: No setup, auto-scaling power.

Cons: Requires stable internet; data privacy considerations.

STEP 4: Path B Direct Download / Local

Best For: Drones, edge devices, and disconnected environments.

Mechanism: Use Git LFS to download model weights and config files directly to the local machine.

Integration: Load into the Transformers or Diffusers library on a laptop or Jetson device using Python.

Pros: Works offline; total control over the inference loop.

Thank you for your attention!

Q&A

